

# Amy Lu

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## EDUCATION

**University of Washington – Seattle WA**

**Jun 2027**

B.S. in Electrical & Computer Engineering, B.S. in Applied Math, B.S. in Informatics – GPA: 3.7/4.0

## SKILLS

- Embedded/Technical: Python, CAN, UART, ESP/STM32, TCP/UDP, RTOS, ROS, FSMs, Git
- Product/Analysis: SQL, MATLAB, Java, Excel, React, HTML/CSS, Financial Modeling, Adobe Suite

## EXPERIENCE

**Vehicle Firmware Engineer – UW Formula Motorsports (FSAE)**

**May 2025 – Present**

- Architecting a real-time wireless telemetry system for an **internationally competing** electric racecar, collecting data via CAN busses, UART (IMU/GPS), and analog sensors. Interfaced data parsing with custom PCB; validated hardware-software integration with CAN bus emulation tools
- Migrating from Raspberry Pi and PCB to **NVIDIA Jetson** as the car's primary data logger, enabling multi-bus CAN acquisition and on-board signal processing in conjunction with **autonomous software**. Automated logging, decoding and offline replay analysis, storing logs in SQLite with live broadcasting to an InfluxDB/Grafana stack
- Evaluating mesh network configurations to improve wireless reliability; conducted empirical tests of antenna performance under dynamic, interference-prone conditions. Performed integration testing across over 30 unique sensor channels and 3 communication buses

**Data Intern – Durkee America**

**Jun 2024 – Sep 2024**

- Co-led development of a machine learning tool to classify and label HVAC technical drawings, automating P&ID recognition and reducing manual sorting effort
- Optimized sales, CRM and inventory systems using Python, SQL, and Excel, cutting manual reporting time by **40%** through process streamlining and AI-driven workflows
- Designed interactive dashboards to track key product and logistics metrics, increasing adoption and reducing inventory response times by 15%

**Administrative Lead – UW Formula Motorsports (FSAE)**

**Jan 2024 – June 2025**

- Directly overseeing **40 administrative members** in team logistics, media, communications
- Built financial and operations dashboards to visualize sponsor ROI, event, and merchandise trends through Shopify drop-shipping storefront, informing **\$220K+** in funding strategy
- Led marketing and media strategy for **1M+ audience**, generating **\$25K in donations** through A/B-tested crowdfunding campaigns, events, Shopify storefront, data-backed by SQL & Meta
- Spearheaded events with 400+ attendees, managing logistics, promotions, and team coordination to boost engagement, resulting in features such as [Fox13 News](#)

**IoT Product Design – Wordplay**

**Sep 2023 – March 2024**

- Real-time sensor data recording for [Wordplay.dev](https://wordplay.dev), a creative coding platform teaching programming through interactive storytelling. Focused on motion-driven interaction into browser-based code
- Led research, design, and integration of intelligent typography and piloted audio feedback for accessibility across UX, product design, engineering members of a 4-person team

**UPCOMING: Chassis Controls Firmware Intern – Tesla (Summer 2026)**

## PROJECTS & LEADERSHIP

**Buildspace Co-Director (Sep 2025 - Present)** – leading hundreds of university students in cseed.co's buildspace for individual software, hardware passion projects in cohort groups of 50 every quarter

**1<sup>st</sup> Place at DubHacks Hackathon (Nov 2024)** – [Interactive AR experience](#) using Snapchat Spectacles, Lens Studio, JavaScript, winning the Pacific Northwest's largest hackathon (12k participants)